# Retrieval and De-duplication

## Retrieval and De-duplication I

Retrieval dispatches the configured query across 4 independent literature engines (arXiv, OpenAlex, Semantic Scholar, and Crossref). Each engine is an isolated adapter exposing a uniform `search(query) -> list[Record]` interface; engines that are keyless — arXiv, OpenAlex (Priem et al. 2022), Crossref (Hendricks et al. 2020), PubMed/Entrez (Sayers et al. 2022), SovietRxiv / RussiaRxiv, ChinaRxiv, Europe PMC, and bioRxiv/medRxiv — need no credentials, while Semantic Scholar (Kinney et al. 2023) uses a key when present. SovietRxiv is a translated archive of Soviet-era scientific preprints sourced from Math-Net.Ru and CyberLeninka (SovietRxiv Project 2026); ChinaRxiv serves translated Chinese preprints from ChinaXiv via the same unified API. Both retain original-language PDFs alongside each translation, and their polite rate-limit pool (300/min vs 30/min anonymous) is activated by an optional

`X-API-Email` header. Europe PMC is a keyless biomedical aggregator covering PubMed, PMC, patents, and preprints in a single search call. bioRxiv/medRxiv share one unified date-window + cursor API; unlike the other engines it is not a free-text search endpoint, so the adapter walks the date window page by page and keeps only records whose title and abstract match every query term client-side. Optional full-text resolution queries Unpaywall (Piwowar et al. 2018) for open-access locations. **Multiple dispatch degrades gracefully**: an engine that is disabled in the configuration, lacks a required key, or cannot reach the network returns a *skipped* status, and the run completes from the remaining engines plus the committed offline corpus.

## Engine Details I

Each engine adapter follows a uniform contract: a module-level API URL constant, a pure `_parse_*` parser function, and a `search_*` entry point with pagination, retry, and graceful error handling. All functions accept an injectable `base_url` parameter for hermetic testing with `pytest-httpserver` — no engine hardcodes its URL inside the function body.

# Engine Details I

| Engine | Rate limit | Pagination | Auth |
|---|---|---|---|
| arXiv | 3s between requests | 100/page, offset | Keyless |
| Semantic Scholar | 1 req/s (unauth.) | 100/page, offset | Optional key |
| OpenAlex | Polite pool (mailto) | 200/page, cursor | Keyless |
| Crossref | Polite pool (mailto) | 1,000/page, offset | Keyless |
| PubMed | NCBI usage policy | retstart/retmax | Keyless |
| SovietRxiv | 30/min (300/min polite) | 1–100/page, cursor | `X-API-Email` |
| ChinaRxiv | 30/min (300/min polite) | 1–100/page, cursor | `X-API-Email` |
| Europe PMC | ~10 req/s (undocumented hard limit) | Up to 1,000/page | Keyless |

# Engine Details I

| Engine | Rate limit | Pagination | Auth |
|---|---|---|---|
| bioRxiv/medRxiv | No documented limit | 100/page fixed, cursor | Keyless |

## Engine Details I

Every new search writes `output/data/retrieval_report.json`, a timestamp-free report that records each attempted, skipped, or failed source with fetched, new-record, and duplicate counts. A zero-result response is therefore distinguishable from a disabled adapter or an HTTP failure. The committed corpus predates that report contract, so this paper intentionally does not reconstruct source-specific counts from the merged corpus.

## Canonical Identifier Hierarchy I

Heterogeneous records are reconciled by a **canonical identifier hierarchy** — DOI > arXiv ID > Semantic Scholar ID > OpenAlex ID > a stable digest of the normalized title. When two records share a canonical identifier they are merged, keeping the version with the most complete metadata (a count of non-None optional fields). The DOI is normalized: case-folded, resolver-prefix stripped, so the same paper returned by two engines under case/format-variant DOIs merges. For this run, 80 records carry DOIs, 12 carry OpenAlex IDs, and 0 carry arXiv IDs. The retrieved fixture contains 80 records. The configured phase union retains 46 unique records; the separate bibliometric analysis uses 70 records. Its dated temporal subset contains 70 records published across 2000–2023. These are distinct artifact scopes, rather than interchangeable denominators.

## Relevance Filtering I

After de-duplication, a relevance filter drops papers whose title and abstract contain none of the configured relevance keywords (exoplanet, atmosphere, transit spectroscopy, JWST, molecular detection, atmospheric composition). Keywords are matched case-insensitively; an empty keyword list is treated as no filter to avoid silently wiping the corpus. A year filter then excludes papers published before the configured start year (2000).